

Secret Key Sharing

QIA 2025 Hackathon Challenge

Overview

In most cryptographic protocols and scenarios, we assume Alice and Bob — two parties separated by physical distance and where one of the goals is to agree on a shared secret. We also assume Eve, who attempts to disrupt the Alice - Bob communication or eavesdrop on it.

In the proposed challenge, Alice and Bob need to establish a secret key (a random string of bits) known only to them, which they will later use to protect further sensitive information. Sending such a key directly through a classical communication channel carries an obvious risk, since Eve could intercept the entire message. To overcome this issue, Alice and Bob communicate with a trusted third node, party, Charlie, who is able to generate and distribute entangled photon pairs, sending one photon to Alice and the other to Bob node. This challenge focuses on exploring how Alice and Bob can use these quantum resources to achieve their communication goals.

Assumptions

1. **Communication channel:** a medium used to transmit information between parties.
2. **Classical communication channel:** a communication channel that transmits information using bits.
3. **Quantum communication channel:** a communication channel that transmits information using qubits (e.g., optical fiber).
4. **Noise:** imperfections in the quantum communication channel that introduce loss, decoherence, or measurement errors in the transmitted qubits.
5. **Entanglement and EPR pairs:** *entanglement* refers to non-classical correlations between quantum systems that cannot be reproduced by any classical or locally realistic model. An *EPR pair* (typically one of the Bell states) is a canonical example of a two-qubit entangled state used to provide Alice and Bob with correlated measurement outcomes suitable for generating a shared secret key.

6. **Alice:** one of the two remote participants. Alice receives quantum states from Charlie, performs local quantum measurements, generates local randomness, and exchanges classical information with Bob through the classical channel. Her goal is to establish a shared secret key with Bob.
7. **Bob:** the second participant. Like Alice, Bob receives quantum states from Charlie, performs local measurements, generates local randomness, and communicates with Alice only through the classical channel. His goal is to obtain the same secret key as Alice.
8. **Charlie:** a third party acting solely as a quantum state source. Charlie prepares entangled photon pairs (EPR pairs) and distributes them by sending one photon of each pair to Alice and the other to Bob. Charlie does not participate in classical communication and does not assist in key generation beyond providing entangled states.
9. **Eve:** the adversary. Eve may observe all classical communication, interfere with the quantum channels, and impersonate Alice, Bob, or Charlie whenever the protocol does not include authentication mechanisms.

The team should assume that Alice and Bob are connected exclusively through a classical communication channel, while Charlie is connected to Alice and Bob only through quantum communication channels, as illustrated in the diagram below.

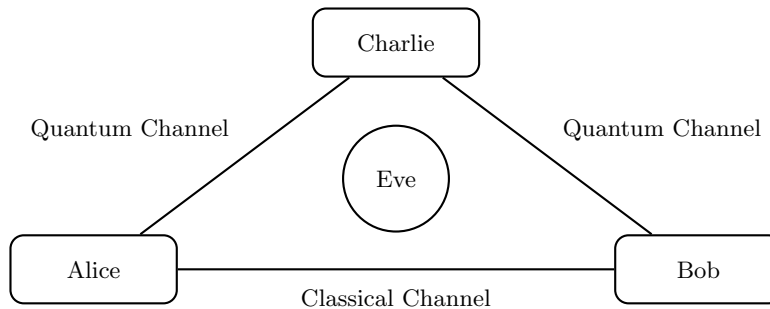


Figure 1: Communication layout for the key-sharing scenario.

Your Task

The team should present a clear list of steps describing how Alice and Bob can use the available resources to obtain a shared secret key while remaining confident that Eve does not learn it. This list should explain what each party does and in what order, starting from the moment entangled states are distributed and ending with Alice and Bob holding matching secret bits.

The team should also identify any stages that may act as bottlenecks, such as points where losses, noise, timing, or coordination difficulties could limit the effectiveness of the approach. In addition, the team should explain how Eve could attempt to interfere with these stages, including ways she might try to disrupt the process, gain information, or impersonate one of the parties.

The final outcome should be a coherent plan that shows how Alice and Bob can successfully obtain a secure shared key and a clear explanation of where the approach is vulnerable and how an adversary might exploit those vulnerabilities. Link monitoring measures should be presented. It should also be described how the services and potential application can be built on to of the proposed scheme - QKD, authentication etc. Any used simulation tools should also be presented and how they optimize the designed scheme.

References

- [1] Davide Rusca and Nicolas Gisin. Quantum cryptography: An overview of quantum key distribution, 2024. arXiv:2411.04044.
- [2] Charles H. Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. In *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, pages 175–179, Bangalore, India, 1984.
- [3] Charles H. Bennett, Gilles Brassard, and N. David Mermin. Quantum cryptography without bell’s theorem. *Physical Review Letters*, 68(5):557–559, 1992.